

E-Commerce Sales Analysis

DATA ANALYSTS & DATA SCIENTISTS



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E- Commerce Sales Analysis

- E-commerce is one of the fastest-growing industries.
- Analyzing sales data helps identify trends, customer preferences, and business opportunities.
- This project focuses on exploring sales patterns and extracting meaningful insights to support decision-making.



Key Objectives: Unlocking Sales Performance

- **Monthly Sales Performance**

Identify peak and lowest sales months to understand seasonality and demand fluctuations.

- **Product Category Analysis**

Determine top-performing and underperforming product categories for strategic inventory management.

- **Sub-Category Deep Dive**

Further refine product insights by analyzing sales at the sub-category level.

Methodology: A Data-Driven Approach

01

Data Extraction & Cleaning

Ensuring data quality and integrity is the first step for accurate analysis.

02

Exploratory Data Analysis (EDA)

Visualizing data to uncover initial patterns, anomalies, and relationships.

03

Feature Engineering

Creating new variables from existing ones to enhance model performance.

04

Statistical Modeling

Applying appropriate statistical techniques to validate hypotheses and predict future trends.

05

Reporting & Visualization

Communicating findings clearly and effectively through interactive dashboards and reports.

Customer Segmentation: Tailoring Strategies



1

Sales by Customer Segment

Understand purchasing behavior and preferences of different customer groups.

2

Profit by Customer Segment

Assess the profitability contributions of each customer segment.

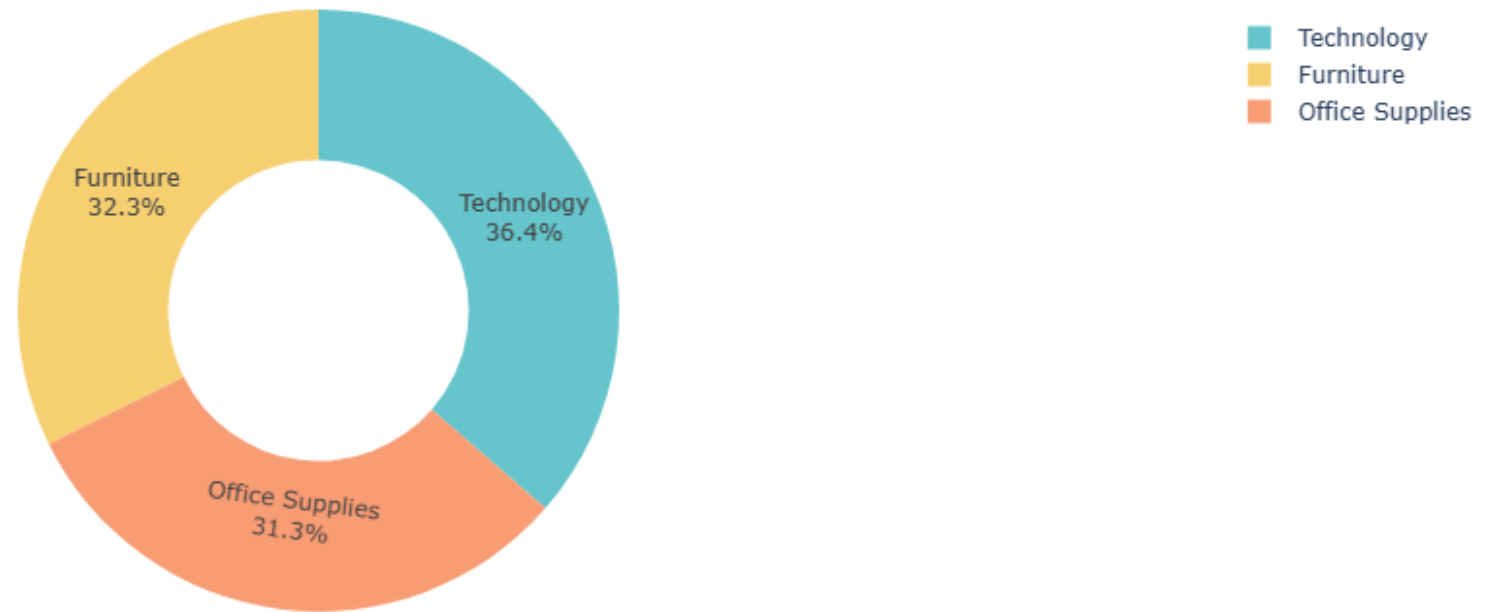
Sales by Month

Monthly Sales Analysis

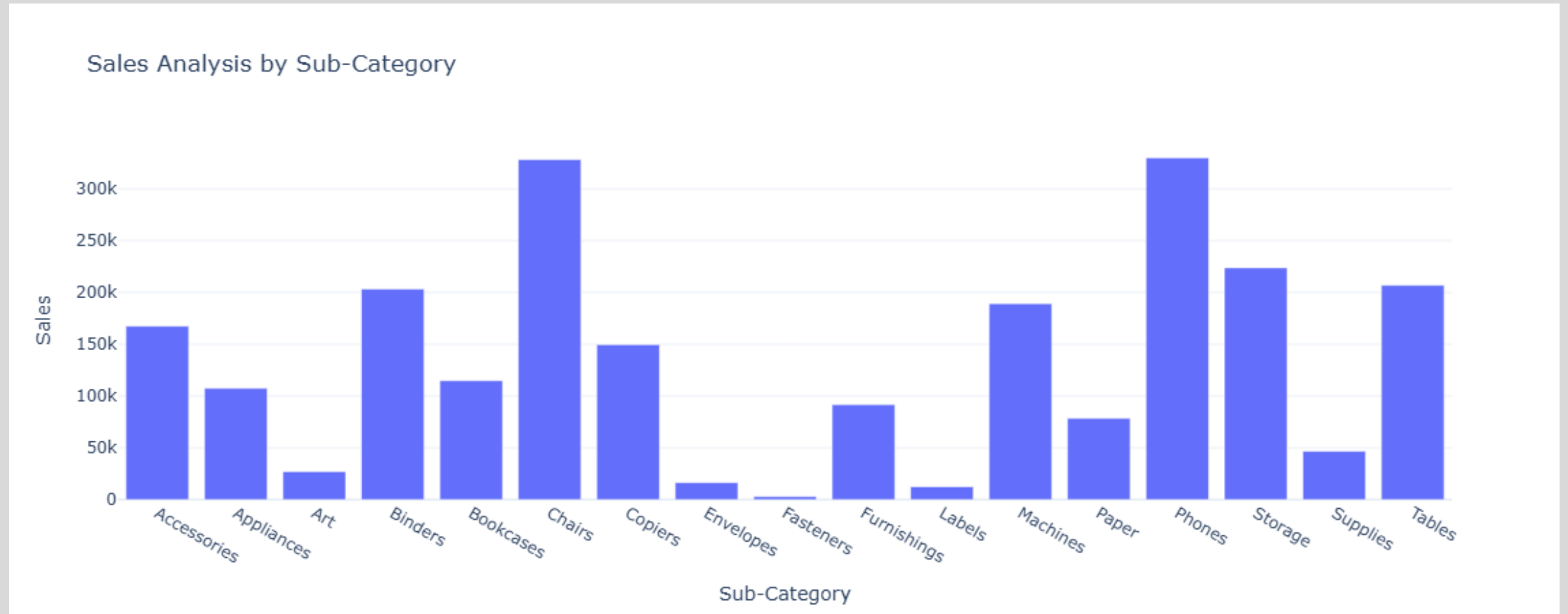


Sales Analysis by Category

Sales Analysis by Category



Sales Analysis by Sub-Category

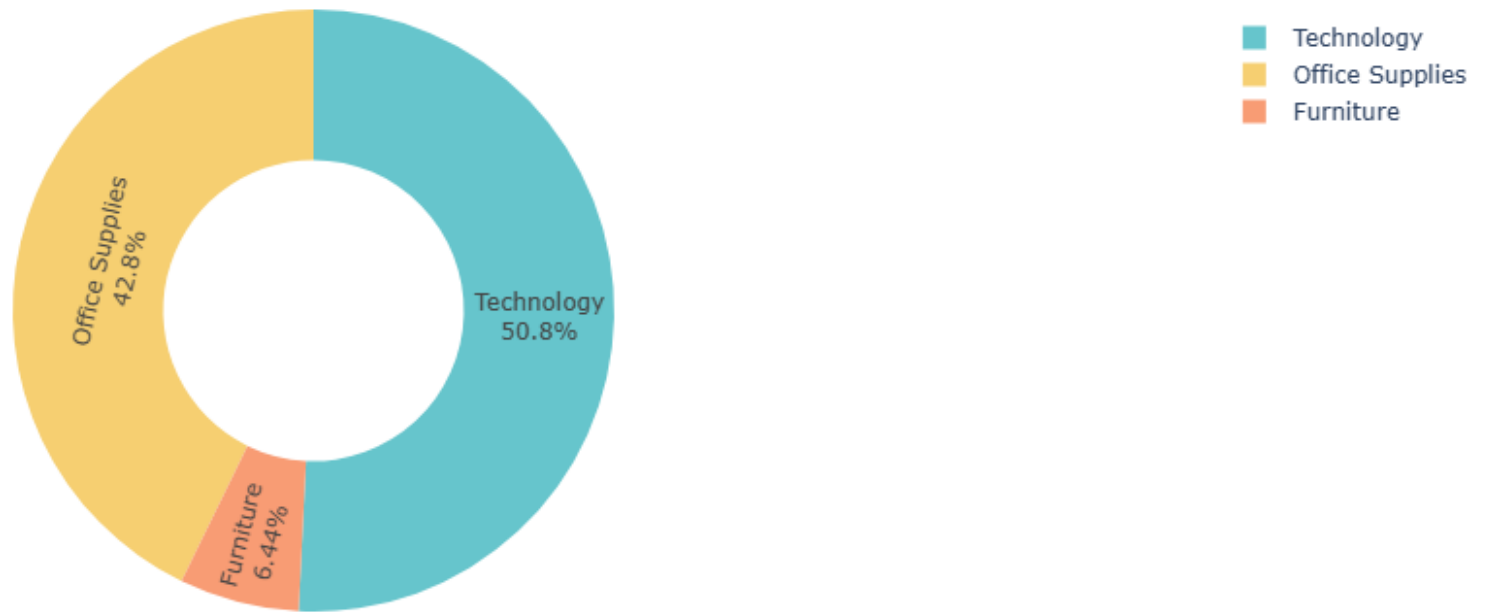


Monthly Profit Analysis

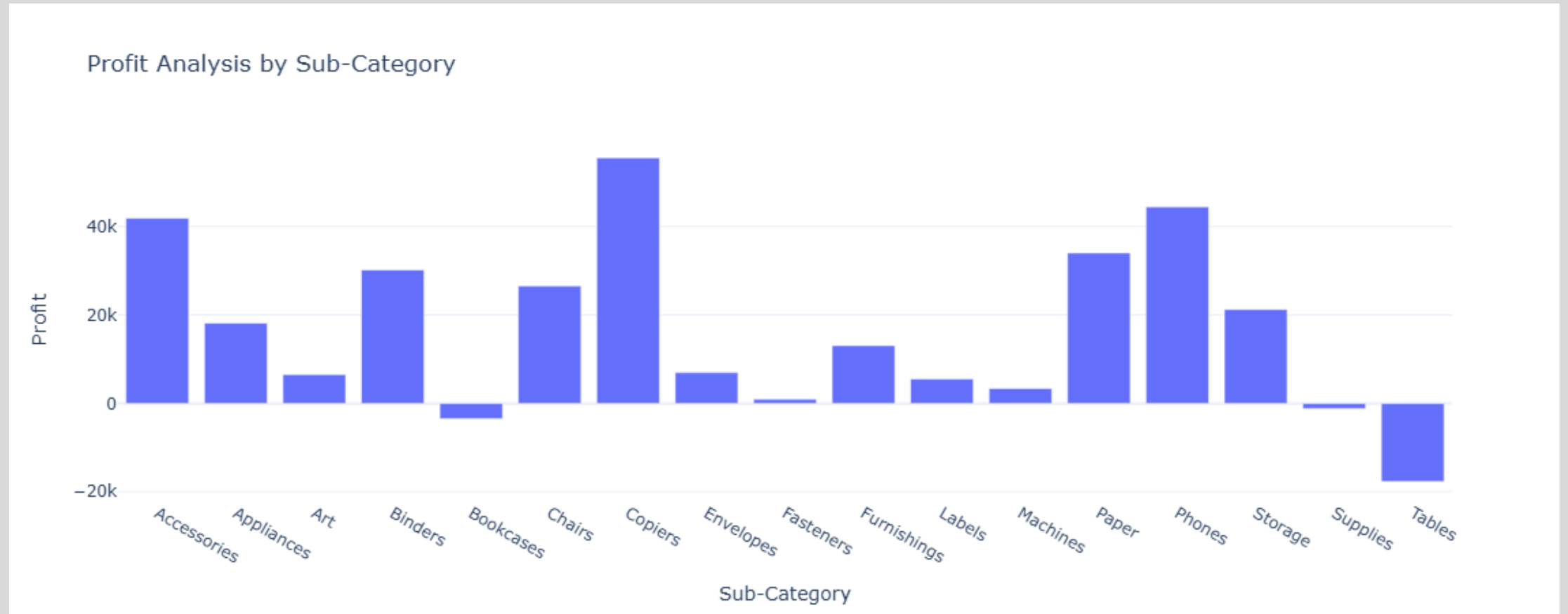


Profit Analysis by Category

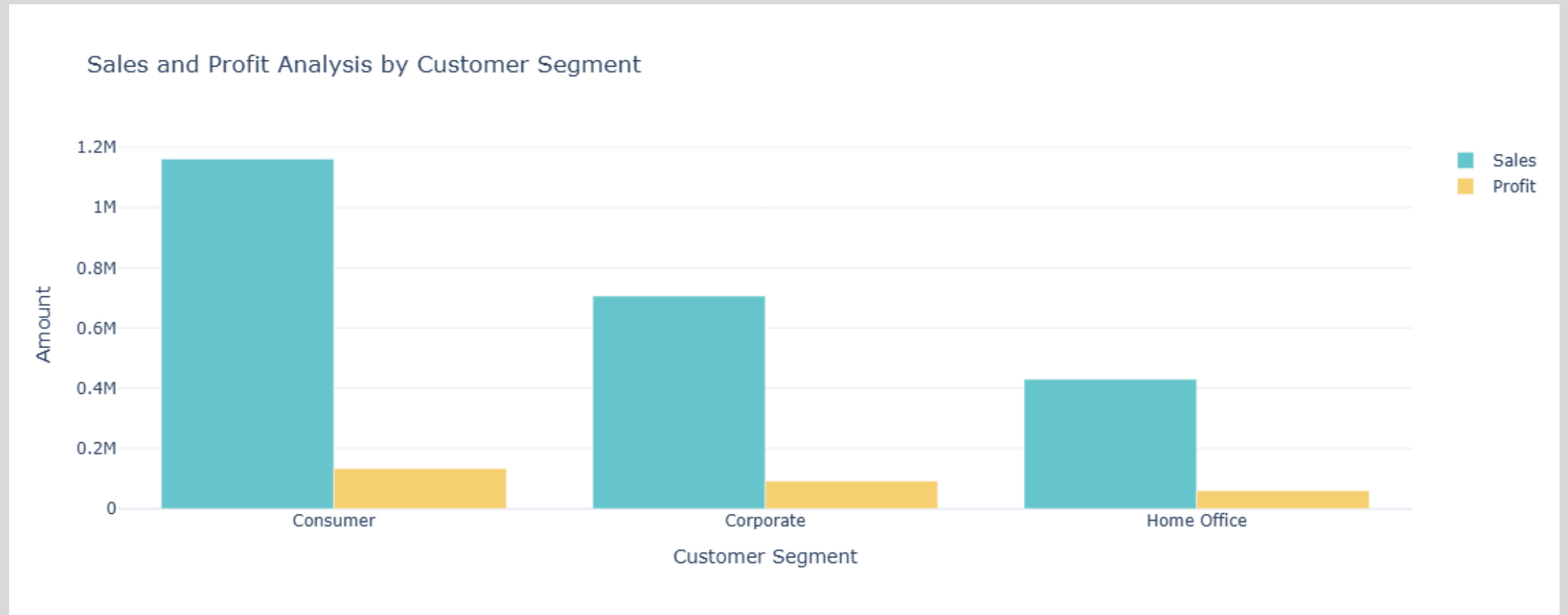
Profit Analysis by Category



Profit Analysis by Sub-Category



Sales and Profit Analysis by Customer Segment



Conclusion

- Data analysis provides deeper insights into customer and sales behavior.
- E-commerce companies can leverage these findings to improve decision-making.
- Continuous monitoring of data will help sustain growth in a competitive market.

